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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 10

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yash Patel**
PRN No.	**23070122235**
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Docker Volumes & Data Persistence
Theory:	Docker volumes provide persistent storage for containers by storing data outside the container's writable layer. Unlike container storage, Docker volumes remain intact even after a container is stopped or removed, allowing data to be shared across multiple container instances. By mounting a named volume to a directory inside the container, applications can read and write data that persists independently of the container lifecycle. This makes Docker volumes ideal for storing application data, logs, databases, and other important files that should not be lost when containers are recreated. **Commands Used** • Built the updated Docker image with a new version tag: docker build -t my-flask-app:v2 . • Created a Docker named volume: docker volume create my-app-data • Started a container with the named volume mounted: docker run -d -p 8082:5001 -v my-app-data:/app/data --name flask-volume-demo my-flask-app:v2 • Stopped the running container: docker stop flask-volume-demo • Removed the stopped container: docker rm flask-volume-demo • Started a new container using the same Docker volume: docker run -d -p 8082:5001 -v my-app-data:/app/data --name flask-

volume-demo2 my-flask-app:v2

Implementation:

Step 1: Modify the Python Application

Modified Flask Application — Updated the Flask application to write timestamped greetings to a log file inside the /app/data directory and configured the application to run on port 5001.

Step 2: Update the Dockerfile

Updated Dockerfile — Modified the Dockerfile to expose port 5001, matching the updated Flask application configuration.

Step 3: Build the New Docker Image

Built Updated Docker Image — Built a new Docker image named my-flask-app:v2 containing the modified Flask application configured for Docker volume-based data persistence.

Step 4: Create a Docker Named Volume

Created Docker Volume — Created a Docker named volume called my-app-data to store application data independently of the container, ensuring data persistence across container restarts and removal.

Step 5: Access the Application and Write Data

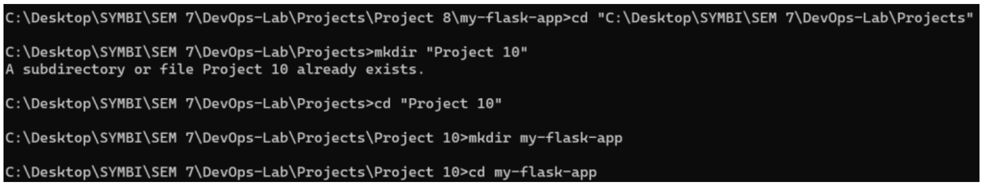
Verified Data Logging — Accessed the Flask application through http://localhost:8082 and refreshed the page multiple times to verify that timestamped greetings were successfully written to the persistent log file stored in the Docker volume.

Step 6: Stop and Remove the Container

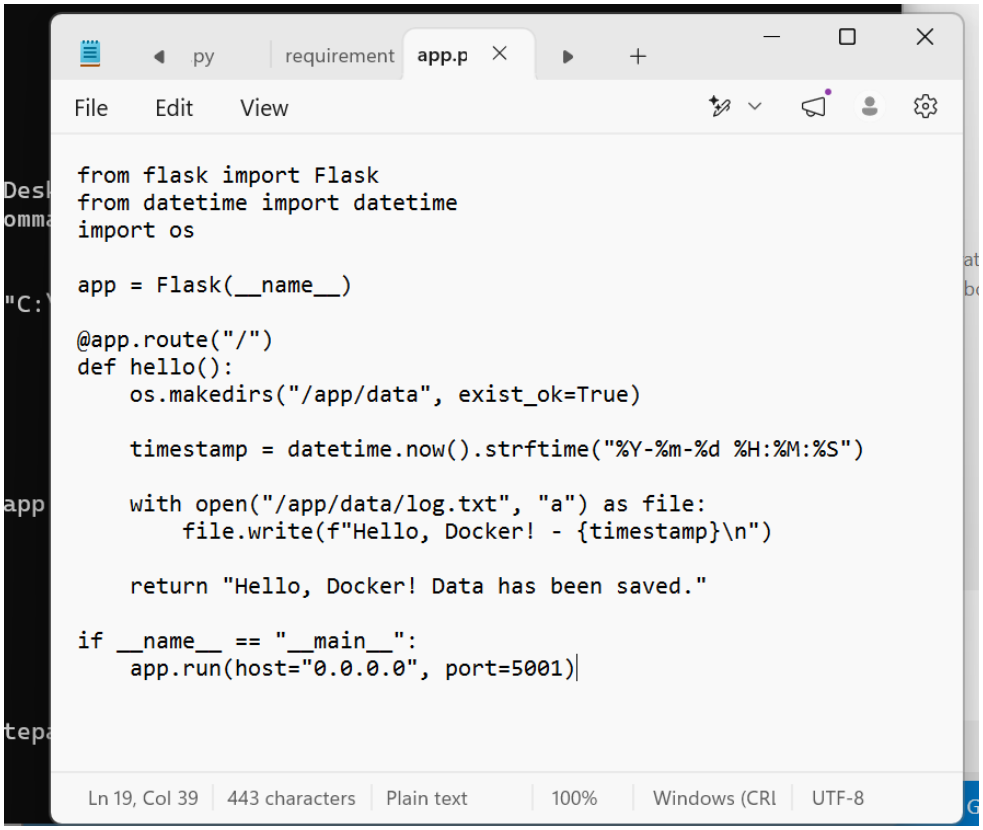
Stopped and Removed Container — Stopped the flask-volume-demo container and removed it using the docker stop and docker rm commands while preserving the application data stored in the Docker named volume.

Step 7: Verify Data Persistence

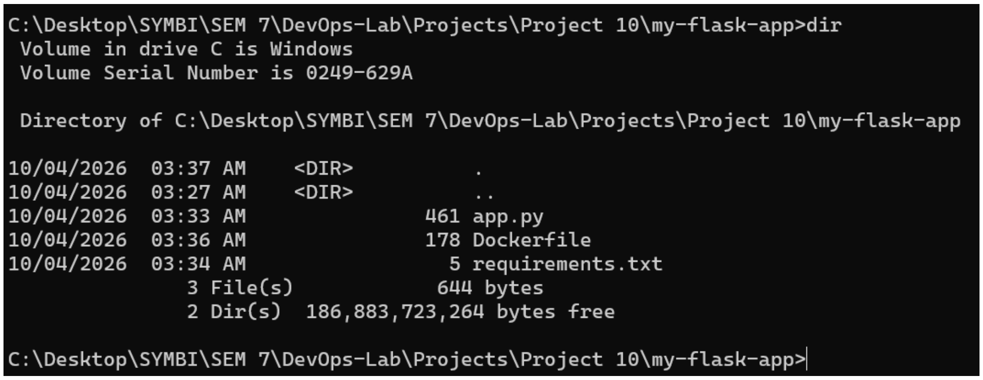
Verified Data Persistence — Accessed the Flask application after starting a new container with the same Docker named volume and confirmed that the previously stored log entries were retained. This demonstrated successful data persistence across multiple container instances.



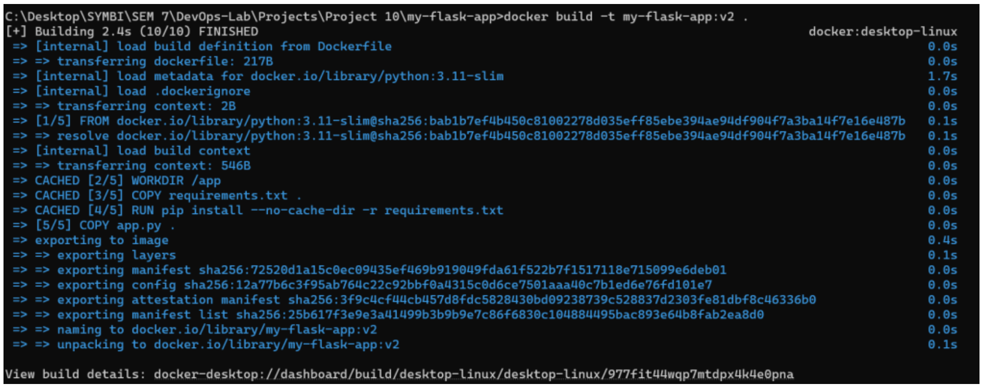
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>mkdir "Project 10"
A subdirectory or file Project 10 already exists.

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>cd "Project 10"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10>mkdir my-flask-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10>cd my-flask-app
```



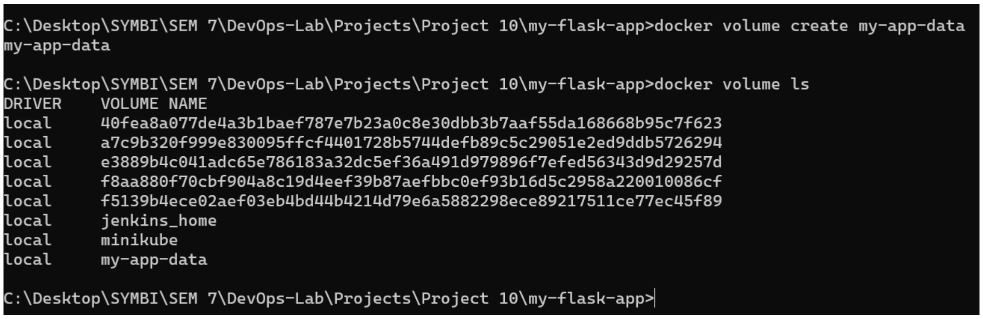
```
from flask import Flask
from datetime import datetime
import os

app = Flask(__name__)

@app.route("/")
def hello():
    os.makedirs("/app/data", exist_ok=True)

    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    with open("/app/data/log.txt", "a") as file:
        file.write(f"Hello, Docker! - {timestamp}\n")

    return "Hello, Docker! Data has been saved."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5001)
```



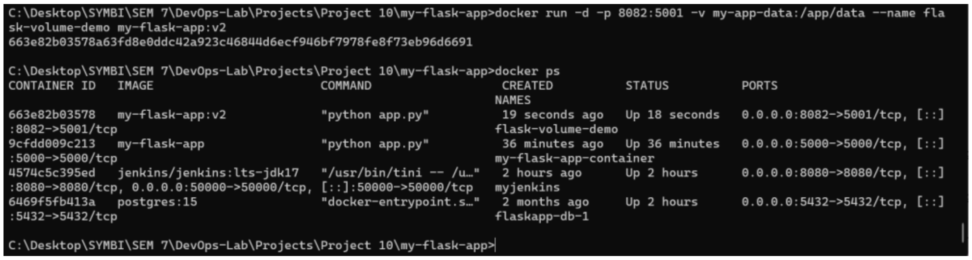
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 0249-629A

 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app

10/04/2026  03:37 AM    <DIR>          .
10/04/2026  03:27 AM    <DIR>          ..
10/04/2026  03:33 AM               461 app.py
10/04/2026  03:36 AM               178 Dockerfile
10/04/2026  03:34 AM                 5 requirements.txt
               3 File(s)            644 bytes
               2 Dir(s)  186,883,723,264 bytes free

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



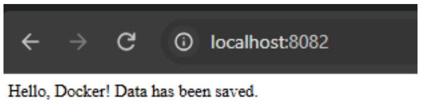
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker build -t my-flask-app:v2 .
[+] Building 2.4s (10/10) FINISHED                                                                         docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 217B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                        1.7s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:bab1b7ef4b450c81002278d035eff85ebe394ae94df904f7a3ba14f7e16e487b  0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:bab1b7ef4b450c81002278d035eff85ebe394ae94df904f7a3ba14f7e16e487b  0.1s
 => [internal] load build context                                                                                          0.0s
 => => transferring context: 546B                                                                                          0.0s
 => CACHED [2/5] WORKDIR /app                                                                                              0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                                   0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                        0.0s
 => [5/5] COPY app.py .                                                                                                    0.0s
 => exporting to image                                                                                                     0.4s
 => => exporting layers                                                                                                    0.1s
 => => exporting manifest sha256:72520d1a15c0ec09435ef469b919049fda61f522b7f1517118e715099e6deb01                          0.0s
 => => exporting config sha256:12a77b6c3f95ab764c22c92bbf0a4315c0d6ce7501aaa40c7b1ed6e76fd101e7                            0.0s
 => => exporting attestation manifest sha256:3f9c4cf44cb457d8fdc5828430bd09238739c528837d2303fe81dbf8c46336b0              0.0s
 => => exporting manifest list sha256:25b617f3e9e3a41499b3b9b9e7c86f6830c104884495bac893e64b8fab2ea8d0                     0.0s
 => => naming to docker.io/library/my-flask-app:v2                                                                         0.0s
 => => unpacking to docker.io/library/my-flask-app:v2                                                                      0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/977fit44wqp7mtdpx4k4e0pna
```



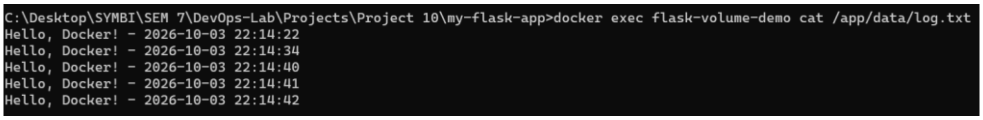
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker volume create my-app-data
my-app-data

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker volume ls
DRIVER    VOLUME NAME
local     40fea8a077de4a3b1baef787e7b23a0c8e30dbb3b7aaf55da168668b95c7f623
local     a7c9b320f999e830095ffcf4401728b5744defb89c5c29051e2ed9ddb5726294
local     e3889b4c041adc65e786183a32dc5ef36a491d979896f7efed56343d9d29257d
local     f8aa880f70cbf904a8c19d4eef39b87aefbbc0ef93b16d5c2958a220010086cf
local     f5139b4ece02aef03eb4bd44b4214d79e6a5882298ece89217511ce77ec45f89
local     jenkins_home
local     minikube
local     my-app-data

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



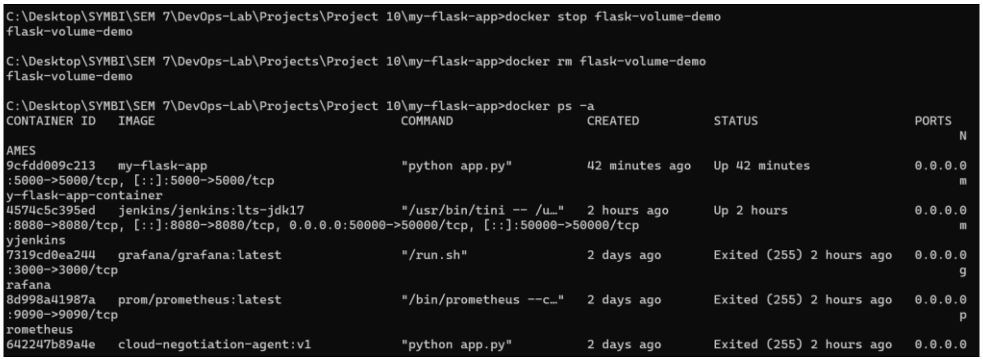
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker run -d -p 8082:5001 -v my-app-data:/app/data --name fla
sk-volume-demo my-flask-app:v2
663e82b03578a63fd8e0ddc42a923c46844d6ecf946bf7978fe8f73eb96d6691

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker ps
CONTAINER ID   IMAGE                       COMMAND                   CREATED          STATUS          PORTS
                                                                    NAMES
663e82b03578   my-flask-app:v2             "python app.py"           19 seconds ago   Up 18 seconds   0.0.0.0:8082->5001/tcp, [::]
:8082->5001/tcp                                                     flask-volume-demo
9cfdd009c213   my-flask-app                "python app.py"           36 minutes ago   Up 36 minutes   0.0.0.0:5000->5000/tcp, [::]
:5000->5000/tcp                                                     my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17   "/usr/bin/tini -- /u…"   2 hours ago      Up 2 hours      0.0.0.0:8080->8080/tcp, [::]
:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
6469f5fb413a   postgres:15                 "docker-entrypoint.s…"   2 months ago     Up 2 hours      0.0.0.0:5432->5432/tcp, [::]
:5432->5432/tcp                                                     flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



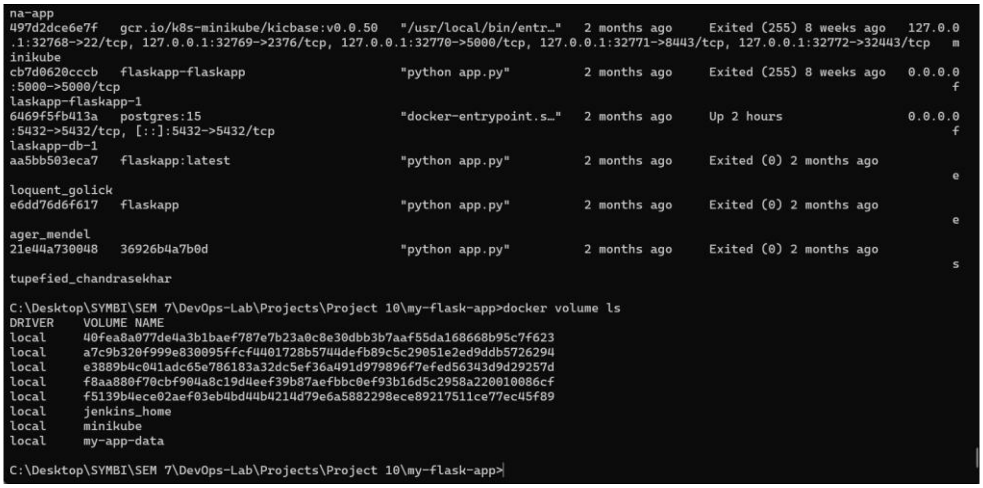
Hello, Docker! Data has been saved.



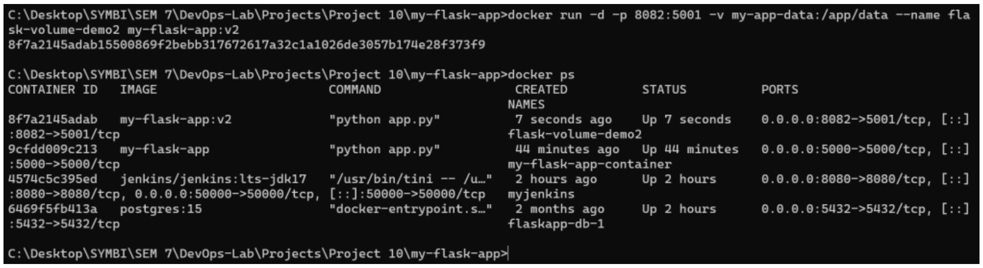
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker exec flask-volume-demo cat /app/data/log.txt
Hello, Docker! - 2026-10-03 22:14:22
Hello, Docker! - 2026-10-03 22:14:34
Hello, Docker! - 2026-10-03 22:14:40
Hello, Docker! - 2026-10-03 22:14:41
Hello, Docker! - 2026-10-03 22:14:42
```



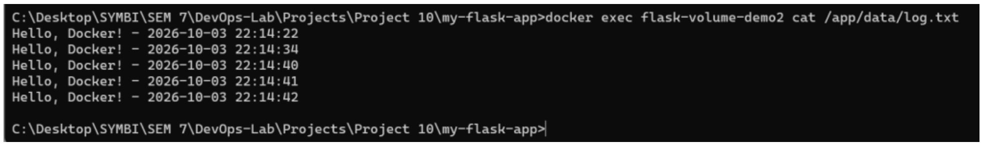
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker stop flask-volume-demo
flask-volume-demo

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker rm flask-volume-demo
flask-volume-demo

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker ps -a
CONTAINER ID   IMAGE                                   COMMAND                   CREATED          STATUS                        PORTS
                                                                                                                                   N
AMES
9cfdd009c213   my-flask-app                            "python app.py"           42 minutes ago   Up 42 minutes                 0.0.0.0
:5000->5000/tcp, [::]:5000->5000/tcp                                                                                               m
y-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17               "/usr/bin/tini -- /u…"   2 hours ago      Up 2 hours                    0.0.0.0
:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp                                              m
yjenkins
7319cd0ea244   grafana/grafana:latest                  "/run.sh"                 2 days ago       Exited (255) 2 hours ago      0.0.0.0
:3000->3000/tcp                                                                                                                    g
rafana
8d998a41987a   prom/prometheus:latest                  "/bin/prometheus --c…"   2 days ago       Exited (255) 2 hours ago      0.0.0.0
:9090->9090/tcp                                                                                                                    p
rometheus
642247b89a4e   cloud-negotiation-agent:v1              "python app.py"           2 days ago       Exited (255) 2 hours ago      0.0.0.0
```



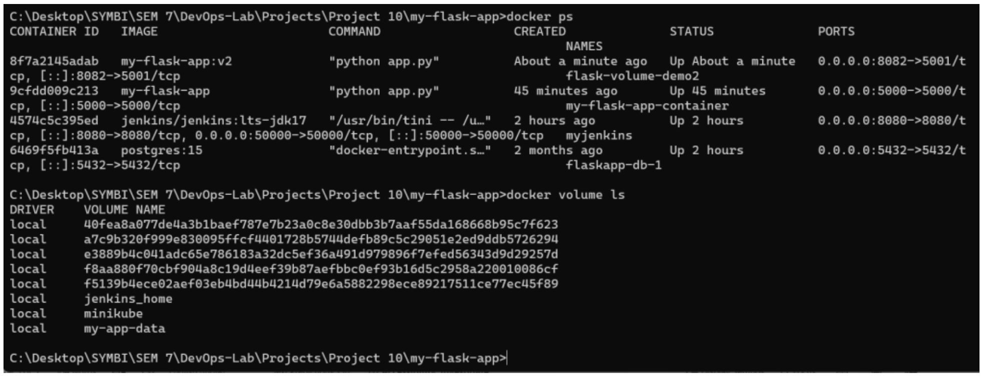
```
na-app
497d2dce6e7f   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   2 months ago     Exited (255) 8 weeks ago   127.0.0
.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   m
inikube
cb7d0620cccb   flaskapp-flaskapp                     "python app.py"          2 months ago     Exited (255) 8 weeks ago   0.0.0.0
:5000->5000/tcp                                                                                                                  f
laskapp-flaskapp-1
6469f5fb413a   postgres:15                           "docker-entrypoint.s…"   2 months ago     Up 2 hours                 0.0.0.0
:5432->5432/tcp, [::]:5432->5432/tcp                                                                                             f
laskapp-db-1
aa5bb503eca7   flaskapp:latest                       "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                                 e
loquent_golick
e6dd76d6f617   flaskapp                              "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                                 e
ager_mendel
21e44a730048   36926b4a7b0d                          "python app.py"          2 months ago     Exited (0) 2 months ago
                                                                                                                                 s
tupefied_chandrasekhar

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker volume ls
DRIVER    VOLUME NAME
local     40fea8a077de4a3b1baef787e7b23a0c8e30dbb3b7aaf55da168668b95c7f623
local     a7c9b320f999e830095ffcf4401728b5744defb89c5c29051e2ed9ddb5726294
local     e3889b4c041adc65e786183a32dc5ef36a491d979896f7efed56343d9d29257d
local     f8aa880f70cbf904a8c19d4eef39b87aefbbc0ef93b16d5c2958a220010086cf
local     f5139b4ece02aef03eb4bd44b4214d79e6a5882298ece89217511ce77ec45f89
local     jenkins_home
local     minikube
local     my-app-data

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker run -d -p 8082:5001 -v my-app-data:/app/data --name fla
sk-volume-demo2 my-flask-app:v2
8f7a2145adab15500869f2bebb317672617a32c1a1026de3057b174e28f373f9

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker ps
CONTAINER ID   IMAGE                         COMMAND                  CREATED          STATUS          PORTS
                                                                       NAMES
8f7a2145adab   my-flask-app:v2               "python app.py"          7 seconds ago    Up 7 seconds    0.0.0.0:8082->5001/tcp, [::]
:8082->5001/tcp                                                        flask-volume-demo2
9cfdd009c213   my-flask-app                  "python app.py"          44 minutes ago   Up 44 minutes   0.0.0.0:5000->5000/tcp, [::]
:5000->5000/tcp                                                        my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17     "/usr/bin/tini -- /u…"   2 hours ago      Up 2 hours      0.0.0.0:8080->8080/tcp, [::]
:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
6469f5fb413a   postgres:15                   "docker-entrypoint.s…"   2 months ago     Up 2 hours      0.0.0.0:5432->5432/tcp, [::]
:5432->5432/tcp                                                        flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker exec flask-volume-demo2 cat /app/data/log.txt
Hello, Docker! - 2026-10-03 22:14:22
Hello, Docker! - 2026-10-03 22:14:34
Hello, Docker! - 2026-10-03 22:14:40
Hello, Docker! - 2026-10-03 22:14:41
Hello, Docker! - 2026-10-03 22:14:42

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker ps
CONTAINER ID   IMAGE                         COMMAND                  CREATED              STATUS                 PORTS
                                                                                NAMES
8f7a2145adab   my-flask-app:v2               "python app.py"          About a minute ago   Up About a minute      0.0.0.0:8082->5001/t
cp, [::]:8082->5001/tcp                                                         flask-volume-demo2
9cfdd009c213   my-flask-app                  "python app.py"          45 minutes ago       Up 45 minutes          0.0.0.0:5000->5000/t
cp, [::]:5000->5000/tcp                                                         my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17     "/usr/bin/tini -- /u…"   2 hours ago          Up 2 hours             0.0.0.0:8080->8080/t
cp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
6469f5fb413a   postgres:15                   "docker-entrypoint.s…"   2 months ago         Up 2 hours             0.0.0.0:5432->5432/t
cp, [::]:5432->5432/tcp                                                         flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>docker volume ls
DRIVER    VOLUME NAME
local     40fea8a077de4a3b1baef787e7b23a0c8e30dbb3b7aaf55da168668b95c7f623
local     a7c9b320f999e830095ffcf4401728b5744defb89c5c29051e2ed9ddb5726294
local     e3889b4c041adc65e786183a32dc5ef36a491d979896f7efed56343d9d29257d
local     f8aa880f70cbf904a8c19d4eef39b87aefbbc0ef93b16d5c2958a220010086cf
local     f5139b4ece02aef03eb4bd44b4214d79e6a5882298ece89217511ce77ec45f89
local     jenkins_home
local     minikube
local     my-app-data

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 10\my-flask-app>
```

Subject Faculty Name: Prof. Arijit Dutta